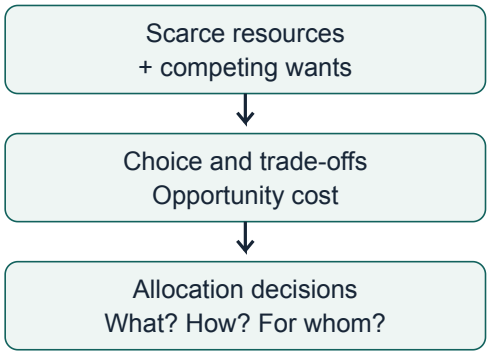
Basic Key Concepts

1. Economics: scarcity → choice → allocation

- **Economics:** studies production, distribution, exchange and consumption; how scarce resources with alternative uses meet competing wants. **Economy:** the actual system of these activities in a country/region.
- **Economic activity:** allocating resources to wants; includes consumption, production, exchange and investment. Income generation alone is not a complete definition.
- **Consumption:** using goods/services to satisfy wants; final consumption by households versus intermediate use in production.
- **Production:** creating or adding utility/value, including services. Inputs: land, labour, capital and entrepreneurship.
- **Exchange:** buying/selling goods or factors. Product pricing concerns goods/services; factor pricing concerns distribution of factor incomes.
- **Factor rewards:** land → rent; labour → wages; capital → interest; entrepreneurship → profit.
- **Investment:** adding capital goods for future production. Gross investment − depreciation = net investment. Autonomous investment does not depend on current income; induced investment responds to income/output. Education and skills build human capital.
- **Economic agents:** households, firms, government and external sector; banks are financial intermediaries.
- **Models:** simplified assumptions and relationships used to explain, predict and analyse policy: demand–supply, circular flow, AD–AS, IS–LM.



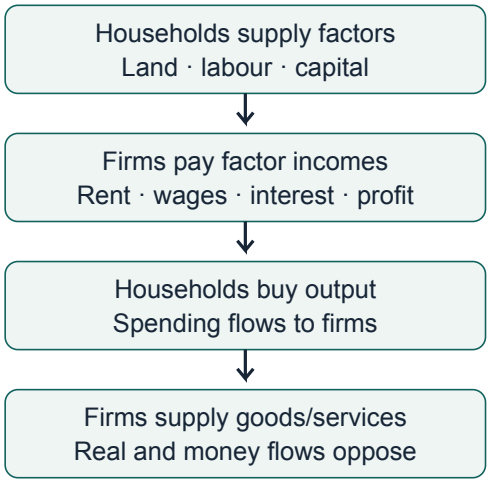
Scarcity forces choices; every choice has a forgone alternative.

- **What/how much?** Consumer versus capital goods; necessities versus luxuries; private versus public provision.
- **How?** Labour-intensive versus capital-intensive techniques.
- **For whom?** Distribution across people (**personal distribution**) and factors (**functional distribution**). Factor shares can also affect inequality.

2. Micro, macro and basic distinctions

- **Microeconomics:** households, firms and individual markets; product/factor prices, demand, production, costs and welfare. Often called price theory.

- **Macroeconomics:** aggregate output, national income, employment, consumption, investment, inflation, growth, trade cycles and income distribution. Uses aggregate demand/supply.
- Micro and macro interact; neither universally assumes every variable in the other branch is constant.
- **Positive:** what is; testable statements about facts/relationships. Example: “A price rise reduces quantity demanded, other things equal.”
- **Normative:** what ought to be; value judgements. Example: “Government should reduce inequality.” Evidence informs these choices but cannot alone settle values.
- **Stock:** measured at a point in time, e.g. capital/wealth on 1 January. **Flow:** measured over an interval, e.g. investment/income per year. A flow changes a stock.
- **Market economy:** decentralised private choices and prices allocate resources. **Planned/non-market economy:** state directs allocation. **Mixed economy:** market choices plus government intervention/social goals; most economies are mixed.
- **Sectors:** primary—agriculture, forestry, fishing, animal husbandry, mining; secondary—manufacturing, construction, utilities; tertiary—transport, trade, banking, tourism and other services. Quaternary—knowledge/R&D/consultancy; quinary—top-level decisions. The last two subdivide services.



Simplified two-sector circular flow; government, finance and external sector add further flows.

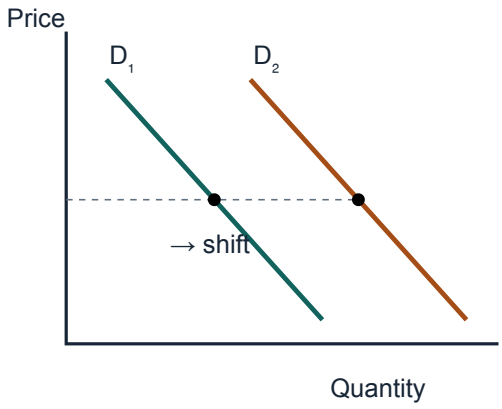
3. Classifying goods

Rival: one person's use reduces availability for others. **Excludable:** access can be restricted.

- **Private:** rival + excludable → food, clothes, cars.
- **Common resource:** rival + non-excludable → open-access fish stocks; overuse risk.
- **Club:** non-rival + excludable → subscription services, an uncongested toll road. Congestion can introduce rivalry.
- **Public:** non-rival + non-excludable → national defence; free-rider problem.
- Government funding does **not** automatically make a good a pure public good. Schools/hospitals can be excludable and congestible; coal/petroleum are not automatically common resources.
- A good can differ by physical attributes, date/time and location.

4. Demand: willingness + purchasing ability

- Demand specifies a **quantity, price and time period**; desire alone is insufficient.
- **Individual demand:** Qd = f(own price, related prices, income, tastes). **Market demand:** horizontal sum of individual quantities at each price.
- Market determinants also include population, income distribution and expectations.
- **Law of demand:** own price ↑ → quantity demanded ↓, other things unchanged. Standard graph: price on vertical axis, quantity on horizontal axis.
- Why downward? Diminishing marginal utility; substitution effect; income effect. Lower prices may bring more buyers into the market.
- **Movement along curve:** own price changes → expansion/contraction of quantity demanded.
- **Shift of curve:** income, preferences, population, expectations or related prices change → increase/decrease in demand.
- **Substitutes:** tea/coffee; price of one ↑ → demand for the other ↑.
- **Complements/joint demand:** bread/butter, pen/ink; price of one ↑ → demand for the other ↓. They need not be used in fixed proportions.
- **Derived demand:** demand for labour/raw materials comes from demand for final output.
- **Composite demand:** one good has several uses, e.g. electricity for lighting/heating/industry.
- **Normal good:** income ↑ → demand ↑. **Inferior good:** income ↑ → demand ↓ over a relevant income range; classification depends on consumers/context.
- **Giffen good:** inferior good whose adverse income effect outweighs substitution effect; own price ↑ → quantity demanded ↑. Not every inferior good is Giffen.
- **Veblen effect:** high price adds prestige/status appeal. Price-quality beliefs can also change perceived desirability.
- Expected shortages/future price rises may raise present demand. These change expectations, so are not clean exceptions with all other factors fixed.



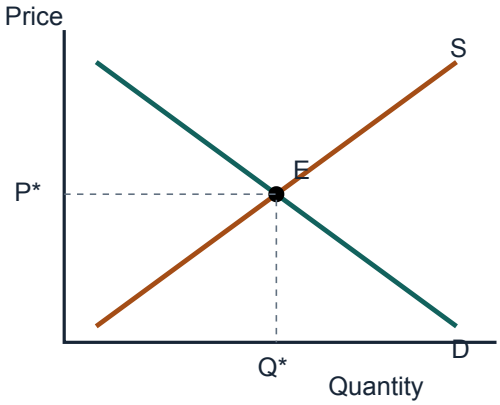
Own-price change moves along D_1. A rightward shift to D_2 means more demanded at the same price.

5. Elasticity: responsiveness, not direction alone

- **Price elasticity of demand:** Ed = % change in Qd / % change in own price. Usually negative; classification normally uses **absolute magnitude |Ed|**. It is dimensionless.
- **Income elasticity:** Ey = % change in Qd / % change in income. Normal: positive; inferior: negative; necessity: 0 < Ey < 1; luxury: Ey > 1.
- **Cross-price elasticity:** Exy = % change in Qx / % change in Py. Substitutes: positive; complements: negative.
- **Five degrees** (demand magnitude or supply elasticity): 0 = perfectly inelastic, vertical; between 0 and 1 = inelastic; 1 = unit elastic; above 1 = elastic; infinity = perfectly elastic, horizontal.
- At perfectly elastic demand, a firm faces a given market price; raising its price loses buyers in the ideal model.
- **Total expenditure/revenue P × Q:** for a price fall, elastic demand → expenditure rises; inelastic → falls; unit elastic → unchanged (local percentage test).
- Demand tends to be more elastic with close substitutes, postponability, more uses, a larger budget share and more adjustment time. Necessities, habits/addiction and dependence on complements can reduce responsiveness. Income/price levels and how narrowly the good is defined also matter; avoid absolute rules.
- **Engel curve:** quantity demanded versus income, holding prices/preferences constant. Normal goods slope upward, inferior goods downward over the relevant range. Necessity/luxury depends on proportional response, not slope alone.

6. Supply and market equilibrium

- **Supply:** quantity sellers are willing and able to offer at a price in a period. A supply schedule lists price–quantity pairs; market supply sums firms' quantities at each price.
- **Law of supply:** own price ↑ → quantity supplied ↑, other things equal.
- **Movement:** own price changes. **Shift:** input costs, technology, alternative products' prices, number of firms, expectations, government policy or producer objectives change.
- Lower input costs/better technology generally shift supply right; higher costs shift it left.
- Harvests, perishables and rare goods may have limited immediate response to prices; this is often a short-run constraint, not a universal reversal of supply law.
- **Supply elasticity:** Es = % change in Qs / % change in price. Same five degrees as above. A horizontal curve means supply at a given price—not “a price increase makes supply zero”.
- Greater spare capacity, accessible/mobile inputs and longer adjustment time generally increase supply elasticity; perishability, capacity limits and technological constraints reduce it.
- **Equilibrium:** Qd = Qs. Above equilibrium price: surplus → downward price pressure. Below: shortage → upward pressure, where prices can adjust.



At E, Qd = Qs. Above P*: surplus. Below P*: shortage.

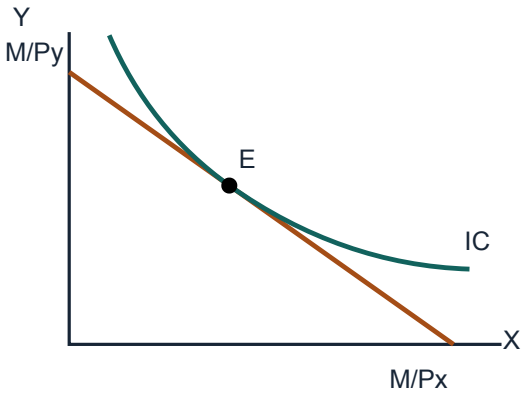
7. Consumer behaviour and utility

- Consumer equilibrium: highest attainable satisfaction within income and prices.
- **Utility:** want-satisfying capacity; subjective, psychological and dependent on person, place and time. It is not a moral judgement. Expected satisfaction guides choices.
- **Cardinal approach:** assumes numerical utility ("utils"); associated with Marshall. **Ordinal approach:** ranks bundles; associated with Hicks and Allen.
- **TU:** total satisfaction. **MU = ΔTU/ΔQ**; for one extra unit, MUn = TUn − TUn−1. TU is the sum of successive MUs (zero baseline).
- Example: TU of 10 units = 150, of 11 = 160 → MU of 11th = 10 utils.
- **Diminishing MU:** successive units add less satisfaction, other things constant. MU positive → TU rises; MU zero → TU stops rising/peaks in the standard model; MU negative → TU falls.
- **Equimarginal principle/Gossen's second law:** spend so that the last rupee on each good gives equal MU: **MUx/Px = MUy/Py = marginal utility of money** (standard interior optimum).
- Single-good equilibrium: **MUx/Px = MUm**; "MU = price" needs utility already expressed in money or MUm normalised to 1.

8. Indifference curves and budget line

- **IC:** bundles of two goods giving equal satisfaction. **IC map:** several ranked curves.
- Standard assumptions: rational, complete/transitive preferences; ordinal ranking; more is preferred; diminishing marginal rate of substitution.
- **MRS = |ΔY/ΔX|:** Y sacrificed for one more X at unchanged utility. It typically falls as X becomes abundant and Y scarce.
- Standard IC: downward sloping, convex to origin; higher curves give more utility; curves do not cross. Perfect substitutes can give straight ICs; perfect complements give L-shaped ICs. "Never touches axes" is not a universal property.
- **Budget line:** PxX + PyY = M; all income spent. Affordable bundles also include points inside the line.
- Intercepts: M/Px on X axis, M/Py on Y axis; slope = −Px/Py.
- Income rises/falls → parallel outward/inward shift, prices fixed. One good becomes cheaper → outward pivot toward that good's axis, other intercept fixed.

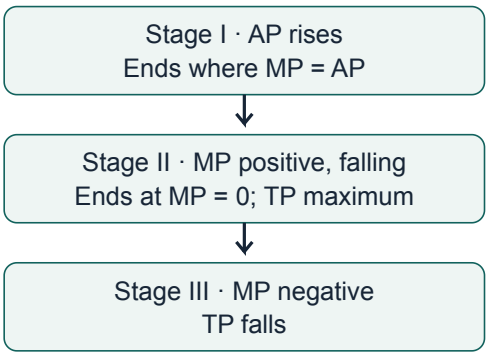
- Standard interior equilibrium: highest reachable IC tangent to budget line, **MRS = Px/Py**. Corner solutions need not satisfy tangency.



Schematic interior optimum: highest feasible IC tangent to budget; MRS = Px/Py.

9. Production, productivity and diminishing returns

- **Production function:** maximum feasible output from inputs at given technology, Q = f(L, K).
- **Short run:** at least one input fixed. **Long run:** all inputs can vary; their proportions need not remain fixed.
- **TP:** total output. **AP = TP/L:** output per unit of variable input. **MP = ΔTP/ΔL:** extra output from extra input. TP = AP × L; with zero baseline TP sums successive MPs.
- MP > AP → AP rises; MP = AP → AP reaches its maximum; MP < AP → AP falls.
- **Law of variable proportions:** vary one input while others are fixed. MP can initially rise, then falls as the fixed input becomes a constraint.
- **Stage I:** AP rises; ends at AP maximum where MP = AP. MP can begin falling before this boundary.
- **Stage II:** MP positive but falling; TP rises at a decreasing rate; ends at MP = 0 and maximum TP. Standard economically relevant stage.
- **Stage III:** MP negative; TP falls.

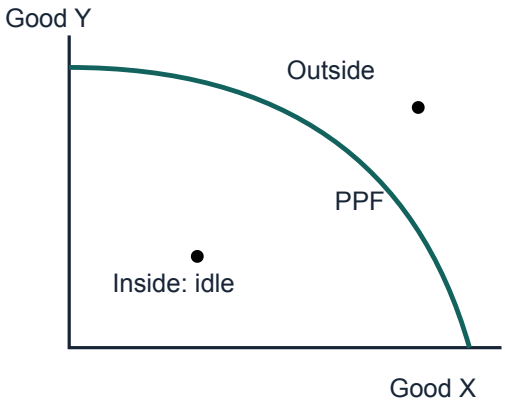


Short run: one variable input, other inputs fixed. Standard production stages.

- **Costs:** input expenditure; **revenue:** sales receipts; **profit = total revenue − total cost**. Economic cost includes opportunity costs.

10. PPF and opportunity cost

- **PPF/PPC:** maximum feasible combinations of two goods with given resources and technology; assumes efficient use of available resources.
- On frontier → productive efficiency; inside → unused/inefficiently used resources; outside → currently unattainable.
- Downward slope shows trade-off. Usual bowed-out shape shows increasing marginal opportunity cost as resources are less suited to the other use. Straight PPF → constant trade-off.
- **MRT/marginal opportunity cost = |ΔY/ΔX|:** Y forgone for extra X.
- **Opportunity cost:** value of the **next-best alternative forgone**. It is not generally "return of forgone option minus return of chosen option".
- ₹2,000 spent instead of invested at 5% for a year → ₹100 interest forgone. State the period when using this comparison.
- A trip costing ₹3,000 plus ₹5,000 earnings forgone → total economic cost ₹8,000: explicit cost ₹3,000 + implicit forgone earnings ₹5,000.
- Growth in resources/broad technology improvement → outward PPF shift; resource destruction → inward shift. Improvement confined to one sector → asymmetric expansion/rotation.
- Moving from inside to frontier improves utilisation; it does not itself shift the frontier.



On PPF: efficient. Outside: unattainable now.
Bowed-out frontier: increasing trade-off.

11. Isocost, isoquant and returns to scale

- **Isocost:** input combinations costing the same: C = wL + rK. Labour intercept C/w; capital intercept C/r; slope −w/r.
- **Isoquant:** input combinations producing the same output; usually downward sloping and convex due to diminishing marginal rate of technical substitution (MRTS).
- Remember: **IC = equal utility; isoquant = equal output; budget = consumer spending; isocost = producer spending.** Standard least-cost tangency: MRTS = w/r.
- **Returns to scale:** change all inputs in the same proportion, technology fixed (long-run experiment).
- **Increasing:** output increases proportionately more; +20% inputs → +25% output. Specialisation/indivisibilities may contribute.
- **Constant:** output and inputs grow in equal proportions.

- **Decreasing:** output increases proportionately less; +15% inputs → +10% output. Coordination difficulties may contribute.
- Physical returns to scale differ from cost economies/diseconomies of scale; the cost link depends on input prices and other conditions.
- **Do not confuse:** diminishing marginal returns changes one input; returns to scale changes all inputs together.

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12. Market structures

A market connects buyers and sellers; it need not be a physical place. Structure depends on numbers, product differentiation, entry barriers, information and factor mobility.

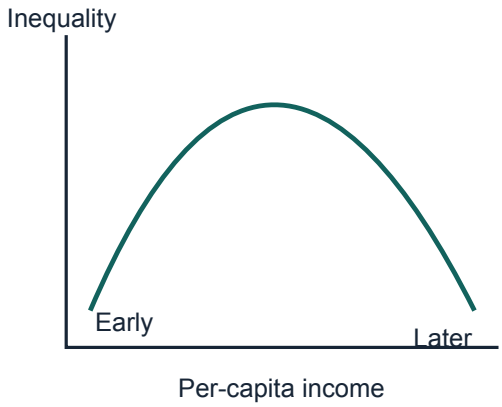
- **Perfect competition:** many buyers/sellers; homogeneous product; free entry/exit; perfect information and mobility; ideal model assumes no transport-cost differences. Firm is a price taker; its demand/AR = price = MR is horizontal. Agricultural markets may approximate some features, not all.
- **Monopoly:** one seller, no close substitutes, strong entry barriers; downward demand/AR, MR below AR. Seller chooses price or quantity subject to demand—not both independently. Price discrimination may occur.
- **Monopolistic competition:** many sellers, differentiated close substitutes, relatively free entry/exit; limited price power; downward firm demand; advertising/selling costs and service-based competition.
- **Oligopoly:** few significant firms; homogeneous or differentiated products; entry barriers; decisions are strategically interdependent. Rival responses to price/output/advertising matter; price matching is not inevitable.
- Oligopoly barriers: large capital, scale, complex technology, patents/licences and regulation. Exit can also be costly.
- **Cartel:** sellers coordinate price/output to act collectively like a monopoly.
- **Duopoly:** two sellers; special case of oligopoly. **Monopsony:** one buyer; buyer power can depress the purchase price when suppliers lack alternatives.
- Monopoly, monopolistic competition and oligopoly are all forms of **imperfect competition**.

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13. Thinkers and economy-wide ideas

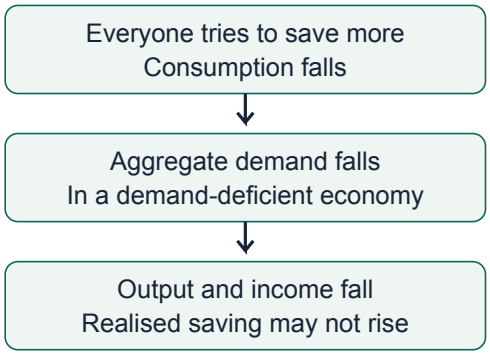
- **Adam Smith (1723–1790):** father of modern economics; *The Theory of Moral Sentiments* (1759), *The Wealth of Nations* (1776); also lectures on jurisprudence.
- **Invisible hand:** self-interest, competition and prices can coordinate economic activity; this is not a guarantee of efficient outcomes with market failures.
- **Laissez-faire:** French for “leave alone”; minimal intervention in market activity.
- Smith's **four tax canons:** equality, certainty, convenience and economy of collection.
- **Lionel Robbins:** scarcity definition; *An Essay on the Nature and Significance of Economic Science* first published **1932**, second edition **1935**.
- **Ragnar Frisch:** introduced micro/macro terminology in 1933.
- **Keynes (1883–1946):** father of modern macroeconomics; Great Depression (1929 onward) motivated aggregate analysis. *The Economic Consequences of the Peace* (1919), *The End of Laissez-Faire* (1926), *The General Theory of Employment, Interest and Money* (1936).

- Positive analysis is associated in the chapter with Say, Senior and Robbins; normative/welfare policy with Marshall, Pigou and Keynes. These are associations, not exclusive categories.
- **Kuznets hypothesis:** development may first raise and later reduce income inequality → inverted U, with per-capita income on X and inequality on Y. It is a hypothesis, not an inevitable law.



Kuznets inverted-U hypothesis: inequality first rises, then falls as development proceeds; not inevitable.

- **Paradox of thrift:** in a demand-deficient economy, everyone trying to save more can cut consumption, aggregate demand, output and income; realised aggregate saving may not rise. This does not mean all saving is harmful; investment and the time horizon matter.
- **Pareto improvement:** someone better off, no one worse off. **Pareto efficiency:** no further such improvement possible. Efficiency does not imply equality or fairness.



Paradox of thrift: an individual intention can have a different aggregate outcome.

14. PYQs in the supplied chapter

- **2021:** Demand may increase when the substitute's price rises or the good's own price falls → **1 and 4 only; option (a)**. A complement's price rise reduces demand; rising income reduces demand for an inferior good. Strictly, own-price fall increases quantity demanded along the curve—the question uses “demand” broadly.
- **2018:** A commodity supplied free by government still has an opportunity cost → **cost transferred to the tax-paying public; option (c)**. A zero consumer price does not erase the cost of scarce resources.

15. Last-minute recall / exam traps

- Own price → **movement**; other determinants → **shift**.
- Inferior ≠ always Giffen; public provision ≠ pure public good.
- Demand elasticity: use magnitude for degrees; sign matters for income/cross elasticity.
- MU is extra **utility**; MP is extra **output**; AP is **TP/input**.
- Equalise **MU per rupee**, not MU alone.
- AP maximum: **MP = AP**; TP maximum: **MP = 0**.
- PPF inside → inefficiency; outside → unattainable now; growth → outward shift.
- Opportunity cost → **next-best forgone alternative**.
- Short-run variable proportions ≠ long-run returns to scale.
- Single seller → monopoly; single buyer → monopsony.
- Pareto efficiency ≠ equitable distribution; Kuznets curve ≠ guaranteed outcome.

Source coverage and correction references

- pp. 1–7 / IMG_5877–5883: foundations, activities, micro/macro, economies, goods, allocation, positive/normative, stocks/flows, sectors.
- pp. 8–13 / IMG_5884–5889: demand, exceptions, elasticities, supply and equilibrium.
- pp. 14–18 / IMG_5890–5894: utility, consumer equilibrium, ICs, budget and Engel curve.
- pp. 18–24 / IMG_5894–5900: production, productivity, stages, PPF, opportunity cost, isoquant/isocost and scale.
- pp. 25–28 / IMG_5901–5904: market structures, thinkers, Kuznets, thrift and Pareto.
- p. 29 / IMG_5905: two PYQs and answers.
- Corrections to utility/MRS checked against NCERT: Consumer Behaviour; productivity against NCERT: Production and Costs.
- Goods classification checked against OpenStax: Public Goods; opportunity cost against OpenStax: Budget Constraints.
- Robbins's publication dates checked against LSE History.